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Symbols for piping acc. to EN ISO 10628 / ISO 14617

Fittings

Globe valve	Spectacle blind
Ball valve	Oriface plate
Gate valve	Flow Restrictor
Flap valve (Butterfly valve)	Silencer
Shut off damper (Guillotine type)	Hose
Shut off damper (Butterfly type)	Compensator, general
Control damper (Butterfly type)	Strainer
Louvre Damper	Filter
Check valve	View Glass
Swing check valve	Spray Nozzle
Condensate drain, general	Injector (AN0000-000)
safety valve, angle pattern, spring loaded	Goggle Valve (hydraulic)
Breather valve	Doubble Flap valve *
Self Actuating Pressure reducing valve	Blowdown Vessel
Self Actuating Pressure/Flow control valve	
Three way valve	
Four way valve	
Needle type	

Valves drivers

Hand-operated manual	
Hand-operated manual, protected against unauthorised actuation	
Piston actuator	
Piston Actuator, Fail Close	
Piston Actuator, Fail Open	
Piston Actuator, Stay last	
Actuating Element with Electromagnet	
Motor drive	
Spring-loaded actuating element	
Diaphragm actuation	
Fitting/valve position in case of power failure	
Valve remains in the last position	
Valve opens	
Valve closes	

Pipe parts

Reducer	
Drain funnel	
Outlet	
Swivel Joint	
Closing plugs	
Flanged Connection	
Blind Flange	
Coupling (general)	
Flow meter	
Insulation	
Insulated and Heated	
Insulated and Heated (electric)	
Flow indicator	
Sight Glass	
ORIFICE TYPE FLOW METER	

Equipment

Pumps; fans; compressors	
Pump (general)	
Centrifugal Pump	
Fan (general)	
Compressor	
Ejector pump	
Screw pump	
Equipment drives	
Electric Motor	
Electric Motor; Variable speed	

Separators; filters	
Separator (general)	
Rotary separators; cyclone Separator	
Electrostatic Precipitaor (ESP)	
Wet Electrostatic Precipitaor	
Filter (gerneral)	
Cartridge Filter	
Bag filter	
Gas filter	
Basket Strainer	

Liftung,conveying and transport equipment	
Belt conveyor (general)	
Chain conveyor (general)	
Chain trough conveyor	
Screw conveyer (general)	
Continuous conveyor (general)	
Switchable continuous conveyor (general)	
Bucket elevator	
Rotary valve	

Mixer; kneader	
Mixer (general)	
Kneader (general)	
Further symbols	
Heat exchanger (general)	
Plate heat exchangers	
Electrical heating unit	
Further symbols	
Forge	
Water seal check valve	

Symbols for Instrumentation

	FIT DIFFERENTIAL PRESSURE FLOW ELEMENT
FIT M	MAG. FLOW TRANSMITTER
	FIT Vortex Flow Element
	MASS FLOW CONTROLLER
	VACUUM BREAKER VALVE

BELT SWITCH

BC	BELT CONVEYOR		
SRL xxx	BELT SWAY SWITCH		
FSL xxx	PULL CHORD SWITCH	PW xxx	WRNING LIGHT
FN xxx	ZERO SPEED SWITCH	PH xxx	WARNING SOUND

PNEUMATIC LEGEND SYMBOLS :

20 Micron, 0-5 BAR	FILTER-REGULATOR-LUBRICATOR UNIT INLET PRE. 6.0 BAR OUTLET PRE. 5.0 BAR (ADJUST.) WITH AUTOMATIC DRAIN
	5/2 WAY, DOUBLE SOLENOID VALVE (24 V DC), WITH LED, MANUAL OVER RIDE, SILENCERS
	2/2 WAY, SINGLE SOLENOID VALVE (24 V DC), WITH LED WITH MANUAL OVER RIDE, NORMALLY CLOSED
	FLOW CONTROL VALVE DN25
	TERMINAL STRIP (20% EXTRA I/O)
	FLOW CONTROL VALVE
	CYLINDER

Example for Pipe Marking

LHF . GNI . 025 . GNL . 0001 . DC3
XXX . XXX . NNN. XXX . 0000 . XXX

- PIPE SPECIFICATION (DC1 TO DC8)
- PROGRESSIVE NUMBER
- FLUID CODE
- PIPE SIZE
- PLANT BREAKDOWN STRUCTURE

Example for Instrument Tagging

F30BA002_SE0001
XNNXXNNN_ XX NNNN

- PROGRESSIVE NUMBER
- INSTRUMENTATION IDENTIFICATION LETTERS
- AREA CODE

Measurements

PI DU100-JP0001	Local
PIRAL DU100-JPIT0001	Control center
PDAL DU100-JDPIT0001	EMSR task is realized with the PLS

Explanation

PIRAL DU100-JPIT0001

- identification letters for instrumentation
- TAG-No.

EPU
XX102_EPUxxxx

EPU, Package unit

Instrumentation Identification Letters acc. to DIN 19227 resp. ISO 3511

First Letters		Addition Letter		Succeding Letters	
A	Analysis *	D	Difference	A	Alarm
B	Burner Flame *	F	Proportion	C	Control
C	Conductivity *	J	Measuring point request	E	Sensing Element
D	Density	Q	Quantity, Integrate, Totalize	H	High Limit
E	Electric Size			I	Indicate
F	Flow Rate			L	Low Limit
G	Position			O	Show Sign
H	Manual Initiated			R	Record
I	Current *			S	Switch, Open Loop Control
J	Power *			T	Transmitter
K	Time			V	Valve Function
L	Level			X	Relay *
M	Humidity			Y	Calculation Function *
N	Noise *			Z	Emergency Release
P	Pressure				
Q	Quality				
R	Radiation				
S	Speed, Rotation, Frequency				
T	Temperature				
U	Multi-Variable				
V	Viscosity Vibration *				
W	Weight, Force				
X	Unclassified Variable, e.g. Video				
Y	Torque *				

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Instrument Air for MHS (LF)
Instrument Air for MHS (LF)

SMS group

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Responsible	Reference Document	Discipline	First angle projection method ISO 128		
SMSA01	–	EA			
Created by	Created on	Scale	Format	Doc. Code	Manufacturing Instructions
DHAN	08.01.2025	1:1	1	PID	SN 200

Doc. Type	Doc. Part	Index	Sheet
DRW	001	A	1/6

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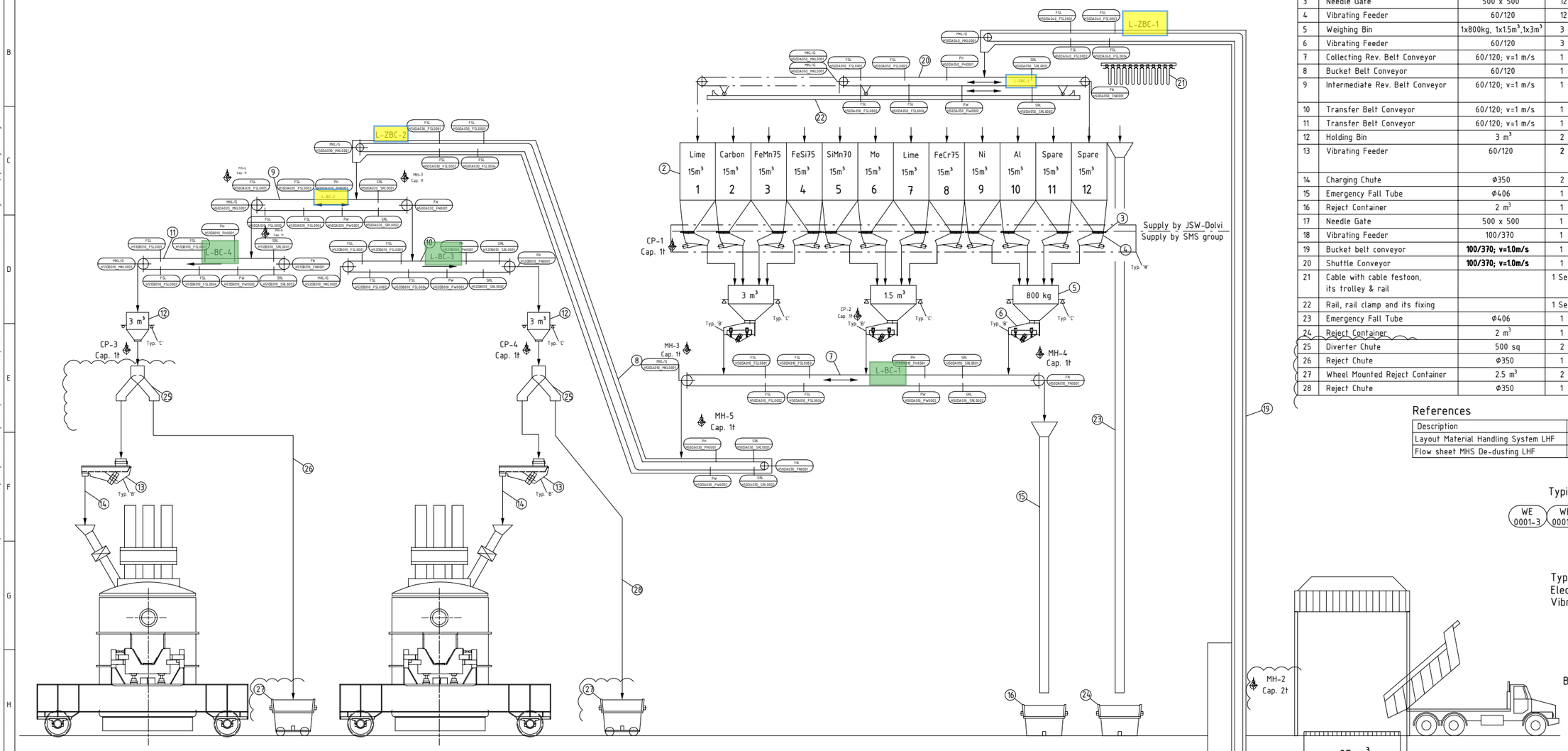

TWIN LF#1

No.	Description	Capacity m³/h, t/h	Qty.	Dimension length/mm, width mm	Remarks	Scope
1	Under ground bunker with canopy	25 m³	1			JSW
2	Storage Hopper	15 m³	12			JSW
3	Needle Gate	500 x 500	12			SMI
4	Vibrating Feeder	60/120	12	approx 1650, 650 wide	magnetic type	SMI
5	Weighing Bin	1x800kg, 1x1.5m³,1x3m³	3			SMI
6	Vibrating Feeder	60/120	3	approx 1500, 650 wide	unbalanced type	SMI
7	Collecting Rev. Belt Conveyor	60/120; v=1 m/s	1	18700 CRS, 650 wide		SMI
8	Bucket Belt Conveyor	60/120	1	19626 CRS., 24280 Lift		JSW
9	Intermediate Rev. Belt Conveyor	60/120; v=1 m/s	1	23500 CRS, 650 wide	along with both discharge chutes	JSW
10	Transfer Belt Conveyor	60/120; v=1 m/s	1	10650 CRS, 650 wide		SMI
11	Transfer Belt Conveyor	60/120; v=1 m/s	1	10650 CRS, 650 wide		SMI
12	Holding Bin	3 m³	2			SMI
13	Vibrating Feeder	60/120	2	3000 CRS, 700 wide	unbalanced type enclosed	SMI
14	Charging Chute	Ø350	2			SMI
15	Emergency Fall Tube	Ø406	1			SMI
16	Reject Container	2 m³	1			SMI
17	Needle Gate	500 x 500	1			**JSW**
18	Vibrating Feeder	100/370	1	approx. 1500, 650	unbalanced type	JSW
19	Bucket belt conveyor	**100/370; v=1.0m/s**	1	12075 CRS, 25375 Lift		JSW
20	Shuttle Conveyor	**100/370; v=1.0m/s**	1	11300, 800 wide		JSW
21	Cable with cable festoon, its trolley & rail		1 Set			JSW
22	Rail, rail clamp and its fixing		1 Set			JSW
23	Emergency Fall Tube	Ø406	1			JSW
24	Reject Container	2 m³	1			JSW
25	Diverter Chute	500 sq	2			JSW
26	Reject Chute	Ø350	1			JSW
27	Wheel Mounted Reject Container	2.5 m³	2			JSW
28	Reject Chute	Ø350	1			JSW

References

Description	Material No.
Layout Material Handling System LHF	16174060
Flow sheet MHS De-dusting LHF	xxxxxxxx

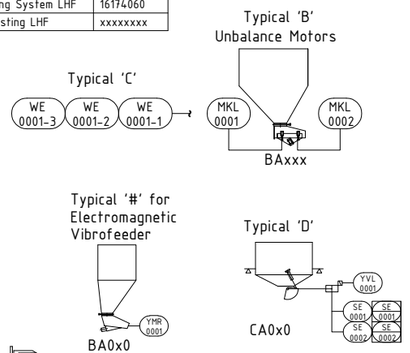

Table of Material Ferro Alloy*

BIN-NO.	MATERIAL	LUMP SIZE (mm)	DENSITY t/m³
01	Lime	10 - 70	1.0
02	Carbon	10 - 50	0.8
03	FeMn75	10 - 50	3.5
04	FeSi75	10 - 50	1.8
05	SiMn70	10 - 50	3.7
06	Mo	10 - 50	2.8
07	Lime	10 - 70	1.0
08	FeCr75	10 - 50	3.7
09	Ni	10 - 50	4
10	Al	10 - 50	1.3
11	Spare	-	-
12	Spare	-	-

'*' Max moisture < 4%, Temp - Ambient, Angle of repose - 35°(assumed) for all Ferro alloys & Lime

List of Hoists for MHS of LF by JSW

NO.	CAPA. (t)	Location	Tentative Level (m)	Lift (m)	Travel (m)
MH1**	1	In Ground Hopper Pit	-0.6	6.0	14.0
MH2**	1	Above Ground Hopper Pit	+4.0	10.0	12.5
MH3	1	Above head end of conv. L-BC-1	+7.0	7.0	7.0
MH4	1	Above tail end of conv. L-BC-1	+7.0	7.0	4.0
MH5**	1	Above tail end of elev. conv. L-ZBC-2	+3.0	3.0	3.5
MH6**	1	Above head end of conv. L-BC-2	+24.0	24.0	3.5
MH7**	1	Above tail end of conv. L-BC-2	+23.35	23.0	5.8
MH8	1	Above tail end of conv. L-BC-4	+20.25	20.0	4.0
CPB 1-4	1	Chain pulley block on various floors for Vibrating Feeder maint.			

'**' Capacities to be checked & confirmed by equipment suppliers.

Legends:

- Weighing system with 3 load cells
- System with one active & two dummy load cell (for full or empty)

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Responsible: SMSA01	Reference Document: -	Discipline: EA	ISO 128 First angle projection method		
Created by: DHAN	Created on: 08.01.2025	Scale: 1:1	Format: 1	Doc. Code: PID	Manufacturing Instructions: SN 200
Doc. Type: DRW	Doc. Part: 001	Index: A	Sheet: 2/6		

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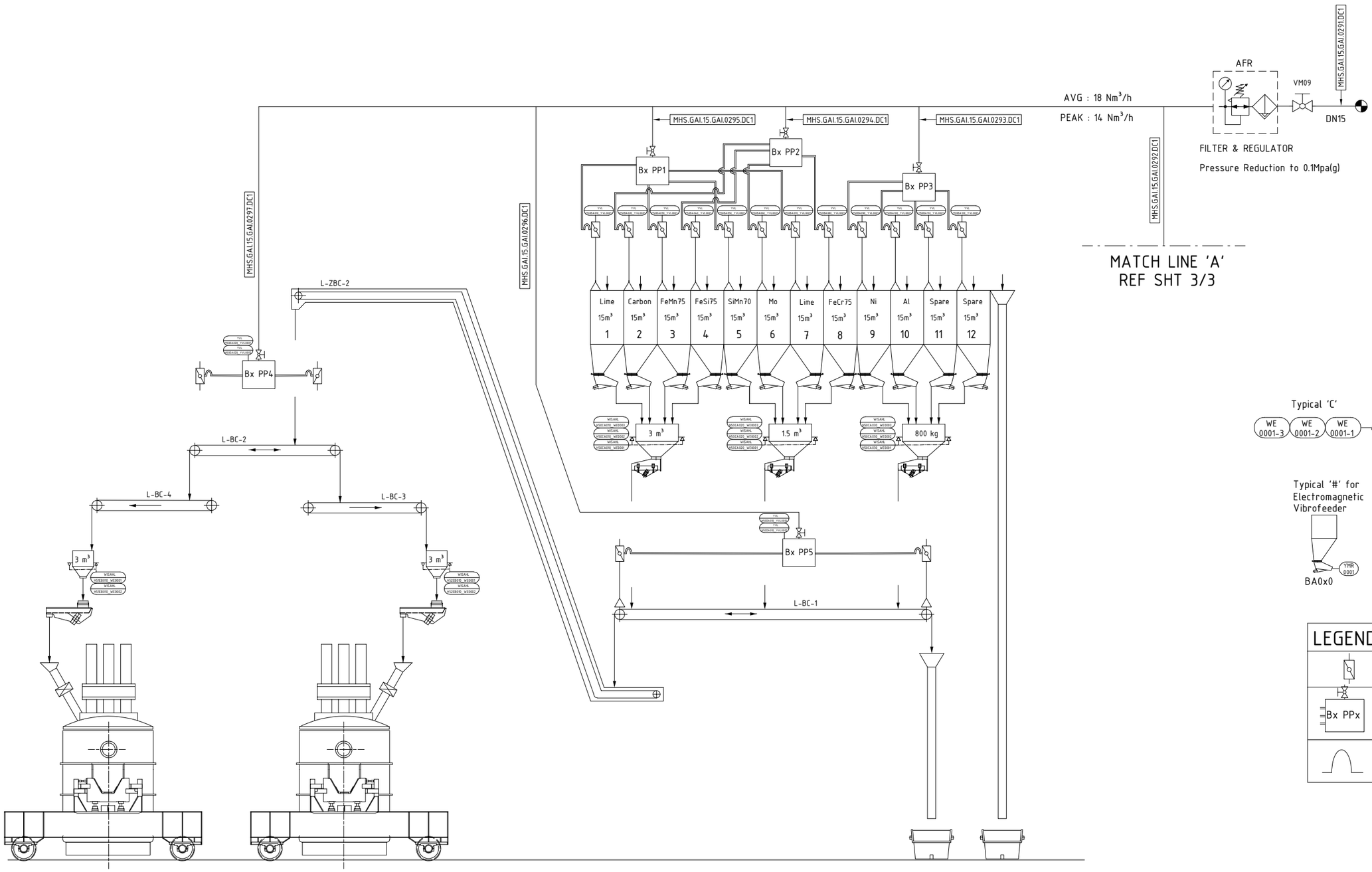
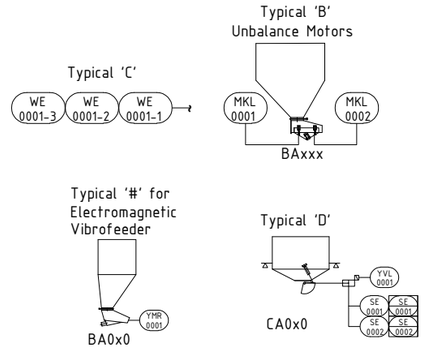


MATCH LINE 'A'
REF SHT 3/3

INSTRUMENT AIR

T.O.P.1 F27 DN 15 (L.F#3)
+15000 mm

Peak	Spitze	18 Nm³/h
Average	Durchschnitt	14 Nm³/h
Pressure	Druck	0.5 - 0.7 MPa (g)



LEGEND :	
	PNEUMATIC DAMPER
Bx PPx	PNEUMATIC PANEL
	HOSE

References

Description	Material
Flowsheet of MHS - Twin LF	16174059
Layout of MHS - Twin LF	16174060
Equipment plan and load data-Twin LF1 MHS	16174061
Flowsheet of MHS De-dusting - Twin LF	16174083

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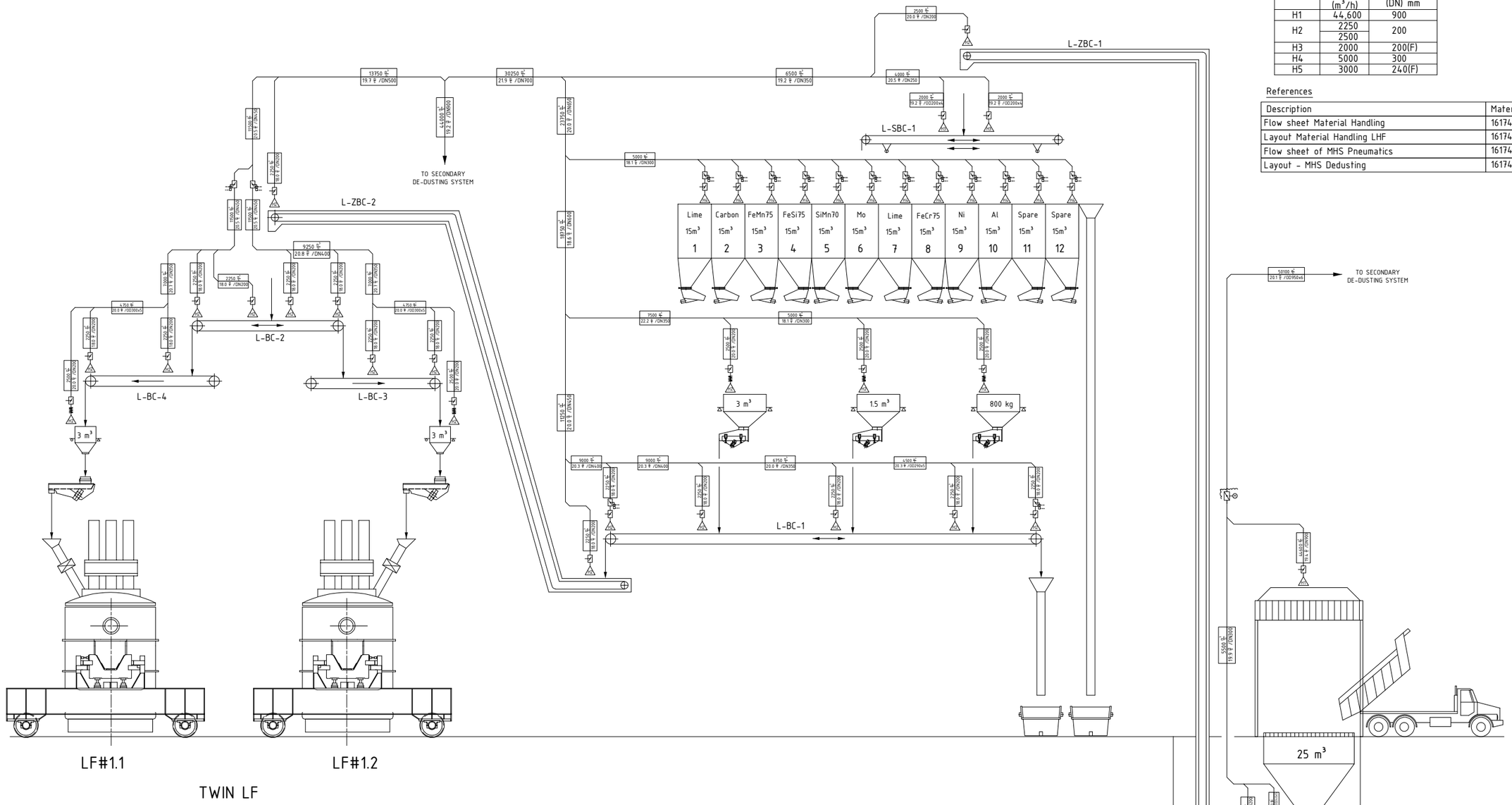

Hood No.	Volume (m^3/h)	DuctDiameter (DN) mm
H1	44,600	900
H2	2250	200
	2500	
H3	2000	200(F)
H4	5000	300
H5	3000	240(F)

References

Description	Material No.
Flow sheet Material Handling	16174059
Layout Material Handling LHF	16174060
Flow sheet of MHS Pneumatics	16174031
Layout - MHS Dedusting	16174084

NOTE-

1. MAXIMUM VELOCITY OF DUST-AIR MIX INSIDE DUCT : 18-22 m/sec.
2. FOR EFFECTIVE SUCTION FROM TRANSFER POINTS ENSURE ALL COVERS/ INSPECTION DOORS ARE IN PLACE & SECURED. EXCESS OPENINGS IN HOODS/ SKIRT TO BE PROPERLY SEALED USING RUBBER STRIP/ CURTAIN.

LEGEND :-

- PNEUMATIC DAMPER ON/ OFF TYPE
- MANUAL DAMPER
- ELECTRIC DAMPER ON/ OFF TYPE
- DUST HOOD WITH HOOD NUMBER
- FLEXIBLE CONNECTION

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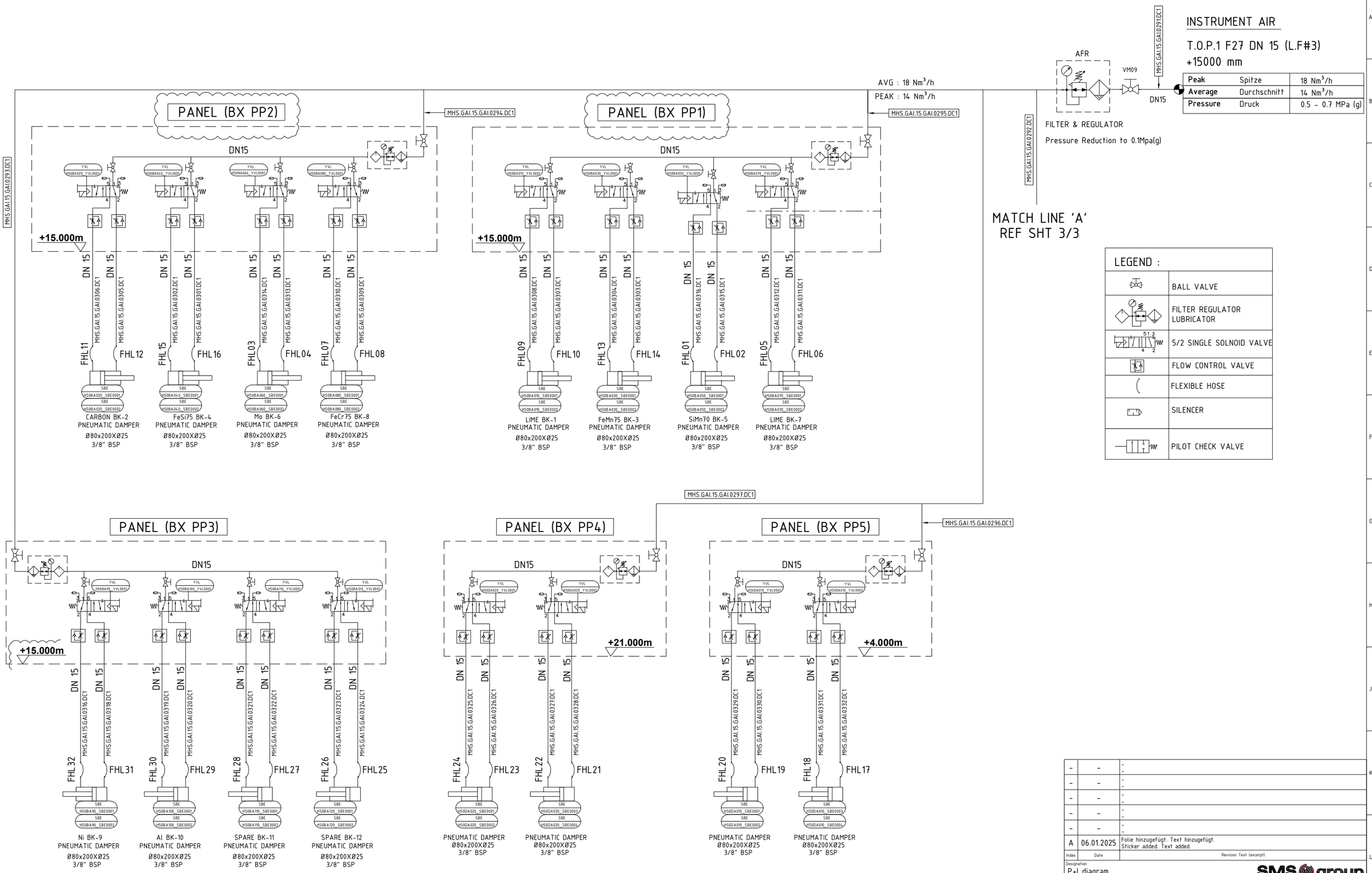

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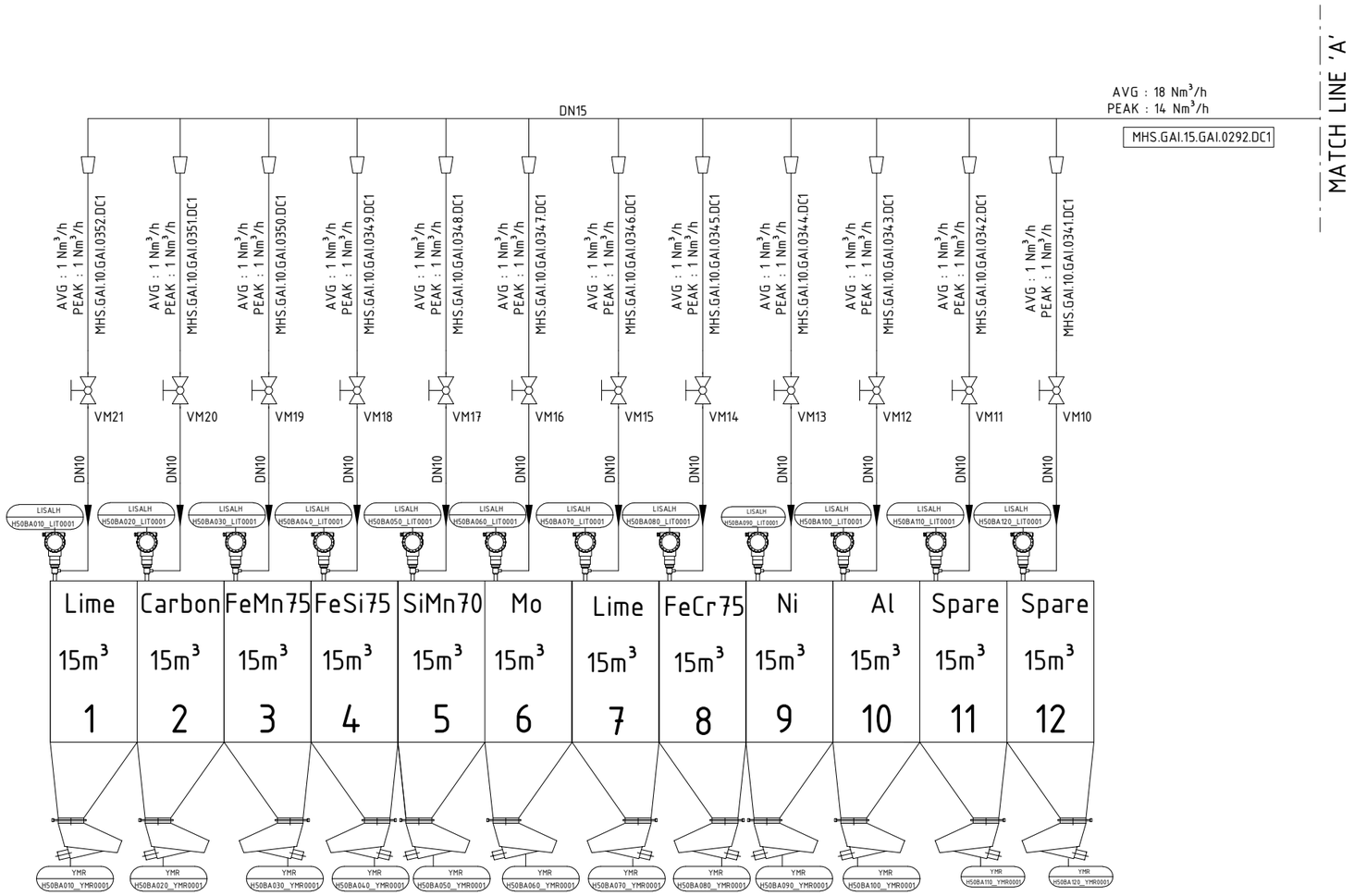

LEVEL SENSOR PURGING SCHEME FOR BUNKERS OF LF

LEGEND :	
	BALL VALVE
	FILTER REGULATOR
	LEVEL SENSOR

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